

PVsyst - Simulation report

Standalone system

Project: Bilhari off grid

Variant: New simulation variant

Standalone system with batteries

System power: 15.30 kWp

3 Storey House with 1200units/permonth - India



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PVsyst V8.1.3

VCO, Simulation date:

06/07/26 07:45

with V8.1.3

Project summary

Geographical Site

3 Storey House with 1200units/permonth

India

Situation

Latitude 23.1261 °(N)

Longitude 79.9661 °(E)

Altitude 396 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

3 Storey House with 1200units/permonth

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Standalone system

Orientation #1

Fixed plane

Tilt/Azimuth 25.5 / 0 °

Standalone system with batteries

User's needs

Daily household consumers

Constant over the year

Average 53 kWh/Day

System information

PV Array

Nb. of modules

34 units

Pnom total

15.30 kWp

Battery pack

Technology

Lead-acid, sealed, Gel

Nb. of units

24 units

Voltage

48 V

Capacity

1248 Ah

Results summary

Simulation step	Hourly				
Useful energy from solar	15917 kWh/year	Specific production	1040 kWh/kWp/year	Perf. Ratio PR	52.56 %
Missing Energy	3245 kWh/year	Available solar energy	25375 kWh/year	Solar Fraction SF	83.16 %
Excess (unused)	8412 kWh/year				

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General parameters

Standalone system

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 25.5 / 0 °

Standalone system with batteries

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

User's needs

Daily household consumers
Constant over the year
Average 53 kWh/Day

PV Array Characteristics

PV module

Manufacturer Generic
Model LG 450 N2W-E6
(Original PVsyst database)
Unit Nom. Power 450 Wp
Number of PV modules 34 units
Nominal (STC) 15.30 kWp
Modules 17 units x 2 In series

At operating cond. (50°C)

Pmpp 14.06 kWp
U mpp 76 V
I mpp 184 A

Total PV power

Nominal (STC) 15 kWp
Total 34 modules
Module area 74.8 m²
Cell area 67.5 m²

Controller

Manufacturer Generic
Model SmartSolar MPPT 150/100 48V
Nb. units 5 units
Technology MPPT converter
Temp coeff. -2.7 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 98.0 / 96.0 %

Battery Management control

Threshold commands as SOC calculation
Charging SOC = 0.90 / 0.75
approx. 55.1 / 50.1 V
Discharging SOC = 0.20 / 0.45
approx. 45.9 / 48.9 V

Battery Storage

Battery

Manufacturer Generic
Model Solar PV 8G8D
Technology Lead-acid, sealed, Gel
Nb. of units 6 in parallel x 4 in series
Discharging min. SOC 20.0 %
Stored energy 47.9 kWh

Battery Pack Characteristics

Voltage 48 V
Nominal Capacity 1248 Ah (C10)
Temperature Fixed 25 °C

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 6.8 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 0.50 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %



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Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, $n(\text{glass})=1.526$, $n(\text{AR})=1.290$

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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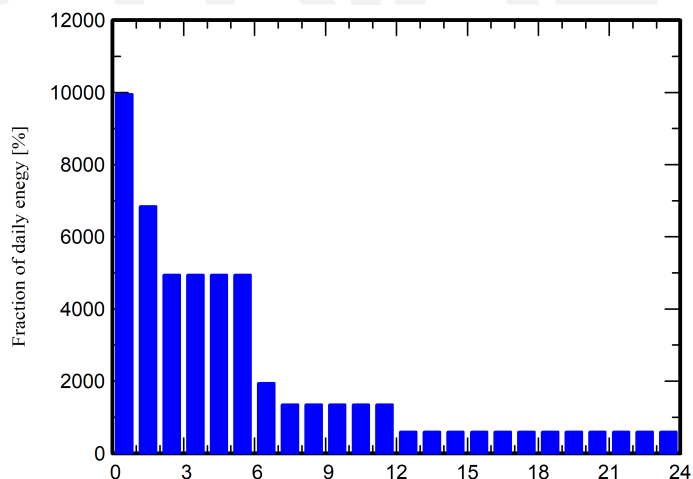
Detailed User's needs

Daily household consumers, Constant over the year, average = 53 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	12	50/lamp	7.0	4200
pump	2	1000/app	1.5	3000
Domestic appliances	1	1200/app	1.0	1200
Fridge / Deep-freeze	2		24	9600
Washing machines/dishwashers	1		2	2700
ac	2	1500 tot	6.0	18000
fans	10	75 tot	12.0	9000
Stand-by consumers			24.0	4800
Total daily energy				52500

Hourly distribution





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Main results

Simulation step

Hourly

System Production

Useful energy from solar 15917 kWh/year
Available solar energy 25375 kWh/year
Excess (unused) 8412 kWh/year

Loss of Load

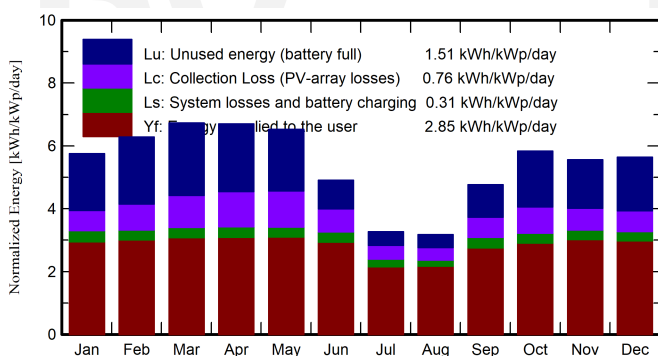
Time Fraction 7.0 %
Missing Energy 3245 kWh/year

Perf. Ratio PR 52.56 %
Solar Fraction SF 83.16 %

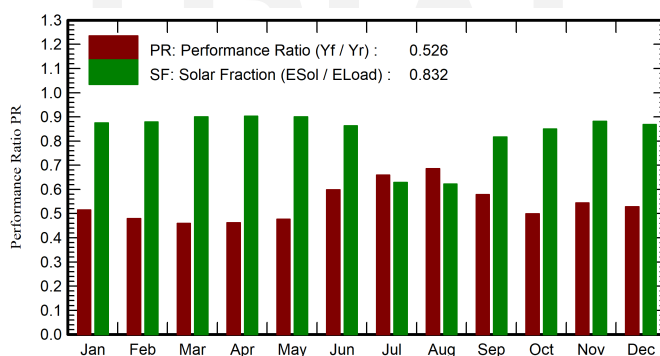
Battery ageing (State of Wear)

Cycles SOW 54.7 %
Static SOW 92.9 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	133.3	175.5	2371	852	224	1403	1628	0.874
February	142.3	173.0	2289	909	179	1291	1470	0.879
March	188.2	204.7	2653	1088	162	1465	1628	0.900
April	201.4	196.8	2507	988	151	1424	1575	0.903
May	218.8	197.5	2490	928	154	1474	1628	0.900
June	163.7	143.4	1860	417	225	1350	1575	0.864
July	111.4	98.6	1285	207	604	1024	1628	0.629
August	104.0	95.9	1260	193	592	1035	1628	0.622
September	137.6	139.8	1837	471	309	1266	1575	0.817
October	155.6	177.5	2315	841	245	1382	1628	0.850
November	130.7	164.4	2189	709	186	1389	1575	0.881
December	127.3	172.1	2317	809	213	1414	1628	0.869
Year	1814.0	1939.1	25375	8412	3245	15917	19163	0.832

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

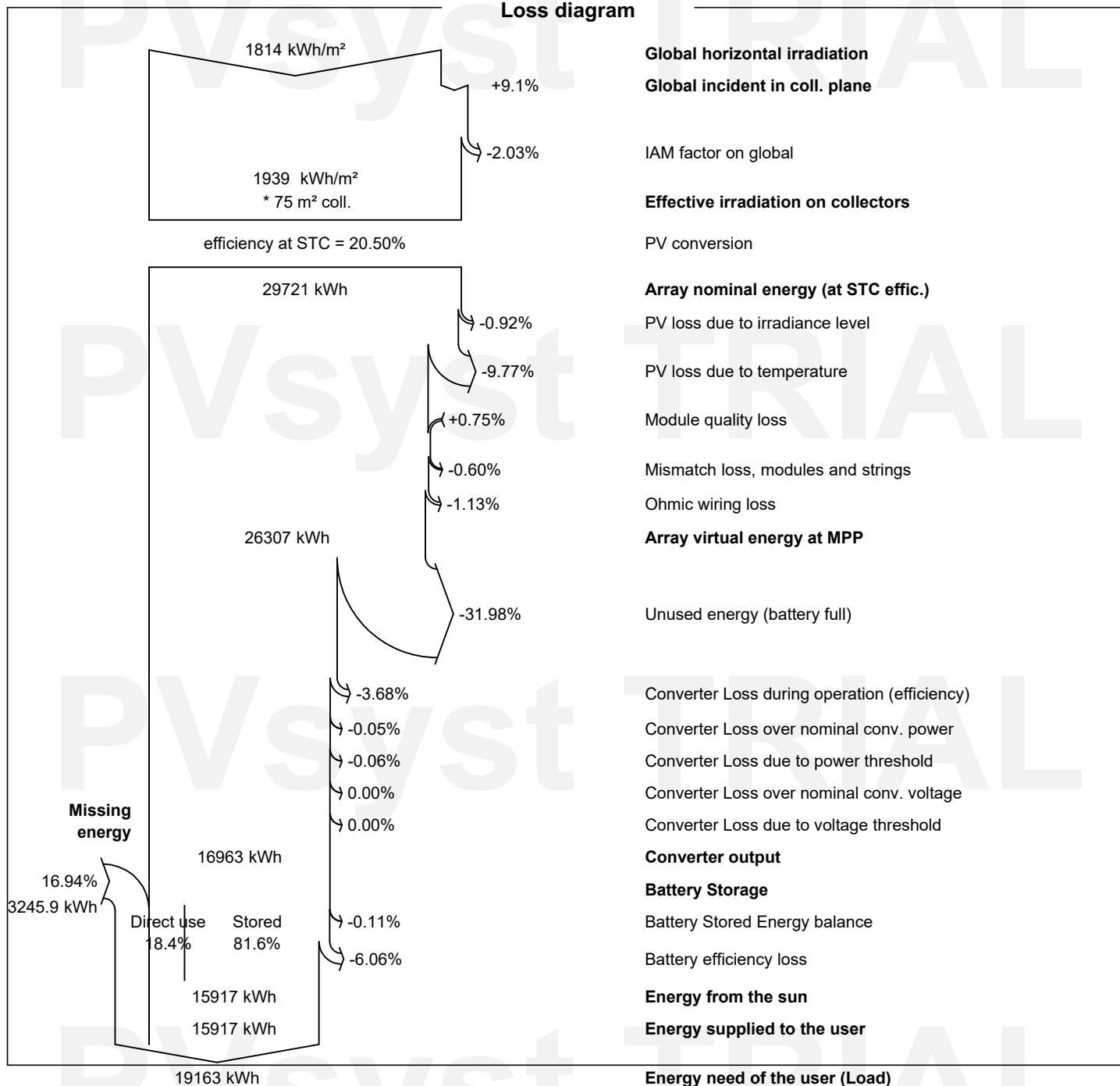
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



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Loss diagram





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Predef. graphs

Daily Input/Output diagram

